

Signal over substance: LLM research judges reward novelty rhetoric, and the bias resists steering

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Abstract

Caveats first. Our judges are three small open models (3.8–8B) and our ideas are *synthetic*; the effect’s magnitude is partly readout-dependent (it attenuates to borderline for one judge in a pairwise readout); and the near-perfect “detectability” of the bias is partly expected because the manipulation is surface-lexical. With that scope stated:

We ask whether an LLM used as a judge for research ideas scores *substantive novelty* (distance from prior work) or *surface novelty-signal* (rhetoric that asserts novelty). We build a correctness-controlled 2×2 design over 154 stems (616 items) crossing substance S (incremental vs. genuine departure, validated by a blind adjudicator) with signal G (plain vs. novelty-rhetoric, same technical content). Across all three judges the signal effect exceeds the substance effect (pooled $\beta_G = 1.03$ vs. $\beta_S = 0.67$), and in 0.82 of matched stems a low-substance idea in novelty rhetoric outscores a genuinely novel idea stated plainly. The effect is *not* explained by length. The signal response is a linear, domain-general direction in the judge’s residual stream (in-distribution AUROC ≈ 1.0 , out-of-domain ≥ 0.997)—yet projecting that direction out, at one layer or many, *does not recalibrate* the judge: β_G survives or β_S collapses with it. Detectability does not imply removability, in contrast to prior activation-steering fixes for self-preference bias. Finally, one mechanism—scoring G not S (signal explains 0.63 of the score variance vs. 0.20 for substance)—reconciles two 2026 benchmarks that separately reported over-rating and score/substance decoupling. If an LLM judge is the verifier in an automated-research loop, it rewards sounding novel over being novel, and the bias cannot be cheaply patched out.

1 Introduction

A recurring thesis holds that the binding constraint on automated AI research is *verification*, not generation: models propose plausible research directions readily, but scoring which are actually good is the bottleneck, and LLM-as-judge is the proposed verifier. A judge is only useful if it rewards real quality. We test the specific failure mode most corrosive to research evaluation: rewarding the *rhetoric* of novelty rather than novelty itself. If a judge does that, then (i) it is a poor verifier, and (ii) placing it in a reinforcement loop trains generators to emit novelty-signalling language, not novel work—reward hacking of an absent oracle.

Contributions. (1) A correctness-controlled 2×2 that isolates surface novelty-signal from substantive novelty (§3); (2) the finding that all three judges score signal over substance, robust to length (§4); (3) a minimal reward-hack demonstrating the score is farmable with substance fixed (§5); (4) a mechanistic result: the signal response is a linear, OOD-generalizing direction that

is nonetheless *not* removable by activation steering (§6)—detectability $\neq$ steerability; and (5) a single-mechanism reconciliation of two 2026 benchmarks (§7).

2 Related work and how we differ

Novelty judging. RQ-Bench [1] finds LLM judges over-rate the novelty of model-generated research questions (a “novelty mirage”); RINoBench [2] finds their novelty verdicts diverge from expert gold with a regression-to-the-mean profile (never “not novel”, rarely top-grade). Both are observational; neither isolates novelty from correctness, nor localizes or intervenes on a mechanism. NovBench [3] argues judges have only surface-level novelty understanding. We supply the controlled crossed design and the mechanism, and reconcile the two benchmarks (§7). **Judge shortcuts.** Judges follow non-quality cues (source, verbosity, confidence, recency) while rationalizing as content [4, 5]; persuasion [6] and self-play [7] inflate scores without correctness. We port the cue-invariance paradigm to *novelty* with correctness fixed by construction, and demonstrate the novelty-specific hack. **Steering judge biases.** “Breaking the Mirror” [8] localizes self-preference bias to a linear direction and *steers it out*; [9] steers judges toward honesty; a creativity direction can be added to amplify creativity [10]. Our probe+direction machinery is the same, but our result is the opposite: the novelty-signal direction is detectable yet *not* steerable away—a dissociation these works did not see.

3 The signal–substance design

Constructs. For each stem (a fixed prior-work context) we author an incremental idea (S =low) and a genuine departure (S =high), matched on plausibility; each is written plainly (G =low) and in novelty rhetoric (G =high) with *identical technical content*. This yields four cells LL, LH, HL, HH. **Validation.** An independent adjudicator, blind to framing, must agree which content departs more (S); all cells must be plausible and balanced with no content leak; and the signalled cells must out-score the plain cells on a judge-independent signal measure (a frozen lexicon and a MiniLM difference-of-centroids). This yields 154 validated stems (616 items) across ML and economics. **Judges & readout.** Phi-3.5-mini, Qwen2.5-7B, Llama-3.1-Nemotron-8B (4-bit). The novelty score Y is the expected digit over the next-token distribution restricted to 1–9 under a frozen rubric that asks for novelty only. We fit $Y \sim S + G + S:G$ with a per-stem random intercept; CIs are stem cluster bootstraps.

4 Judges score signal over substance

Table 1 and Fig. 1: for every judge $\beta_G > \beta_S$ with CIs excluding zero. Pooled $\beta_S = 0.67$, $\beta_G = 1.03$ (ratio 0.65). The *hackability index* $P(Y_{LH} > Y_{HL})$ —a low-substance signalled idea beating a high-substance plain one—is 0.82 pooled (0.89/0.85/0.71 per judge). Judges do track substance ($\beta_S > 0$), but signal dominates.

Not a length artifact (E1). Signalled cells are only $\sim 6\%$ longer; controlling for length within stem, β_G does not shrink (change $-3/-4/-6\%$; the length coefficient is near zero/negative).

Survives comparison (E1). In a pairwise readout with position bias controlled, $P(\text{pick signalled non-novel})$ is 0.70 (Phi) and 0.55 (Nemotron)—both above chance—but 0.53 for Qwen (CI 0.48–0.58, border-

Judge	β_S	β_G	ratio S/G	hackability
Phi-3.5-mini	0.40	0.75	0.54	0.89
Qwen2.5-7B	1.14	1.49	0.76	0.85
Nemotron-8B	0.47	0.84	0.55	0.71
Pooled	0.67	1.03	0.65	0.82

Table 1: E0. Signal effect β_G exceeds substance effect β_S for all judges.

LLM research judges score novelty SIGNAL over SUBSTANCE (N=154 stems, 616 items)

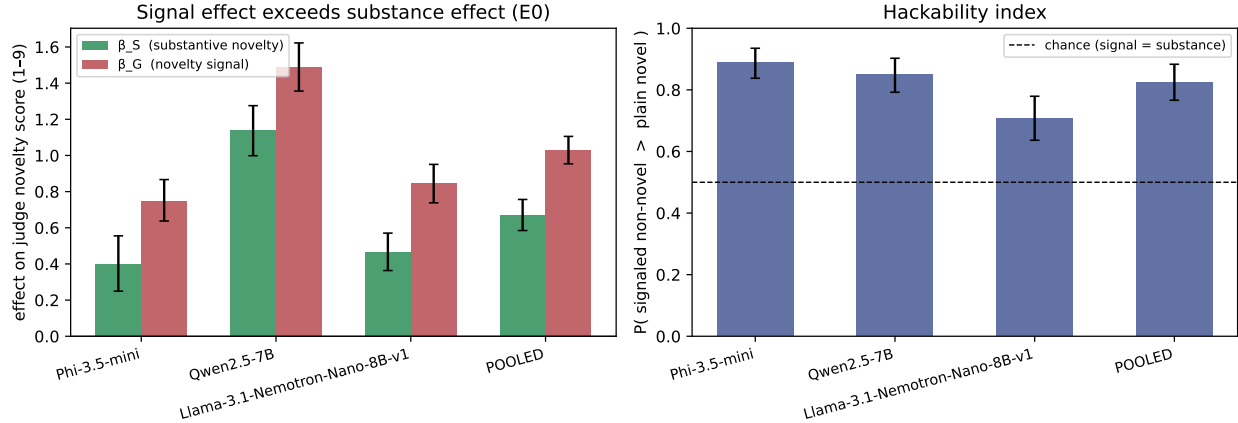


Figure 1: E0: signal beats substance (left) and the hackability index (right).

line). So the effect is not merely a pointwise-readout artifact, though its magnitude is readout-dependent.

5 The score is reward-hackable

A minimal attack—greedily prepending pure-rhetoric sentences (adding no technical content) to a *plain* idea and keeping edits that raise Y —farms the score with substance fixed. On low-substance ideas the gamed gain is 0.99/2.10/1.92 points, and the fraction of low-substance ideas farmed to exceed the judge’s mean score on genuinely high-substance plain ideas rises from 47/7/20% to 100/83/100%.

6 Detectable but not steerable

A linear signal direction (E3). A logistic probe on the scoring-position residual separates G -high from G -low at in-distribution AUROC ≥ 1.000 for all judges, and *across domains* (train ML, test econ) at AUROC ≥ 0.997 . (Near-perfect separability is partly expected: the signal is surface-lexical.) **The fix fails (E4).** Projecting the difference-of-means signal direction out of the residual—at the best probe layer, or across an early-mid band—does not selectively remove the bias (Fig. 2): for Phi β_G is untouched (0.75 $\rightarrow$ 0.77) while β_S falls (0.40 $\rightarrow$ 0.19); for Nemotron both collapse together (β_G 0.84 $\rightarrow$ 0.31, β_S 0.47 $\rightarrow$ 0.13); for Qwen the change is negligible. No judge shows the target ($\beta_G \downarrow$, β_S preserved). The novelty-signal response is entangled with novelty

computation, not a separable direction—unlike self-preference bias, which prior work steered out.

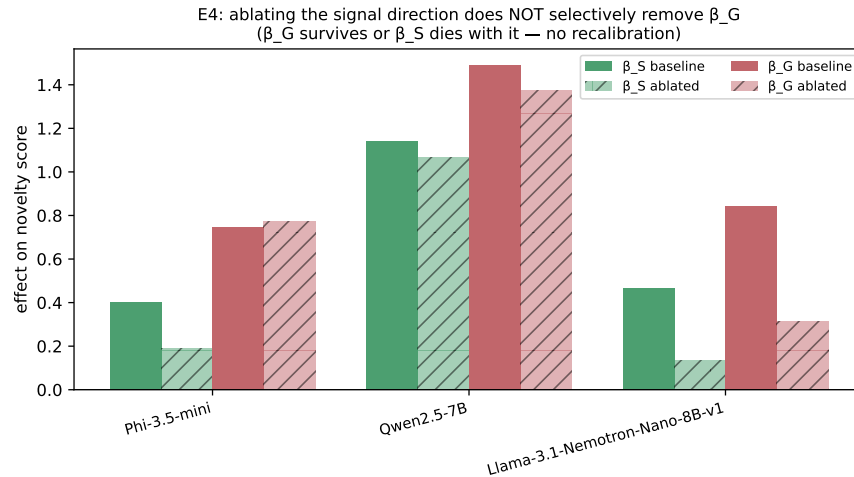


Figure 2: E4: ablating the signal direction does not selectively remove β_G .

7 One mechanism, both benchmarks

If the judge scores G not S , both prior findings follow. The score’s variance is dominated by signal (pooled share 0.63 for G vs. 0.20 for S ; per judge G 0.65/0.46/0.76). Non-novel ideas out-score the average novel idea 0.57/0.33/0.31 of the time (the over-rating of [1]), while genuine novelty framed plainly is under-credited (the decoupling of [2]). External validation on those corpora is future work (Fig. 3).

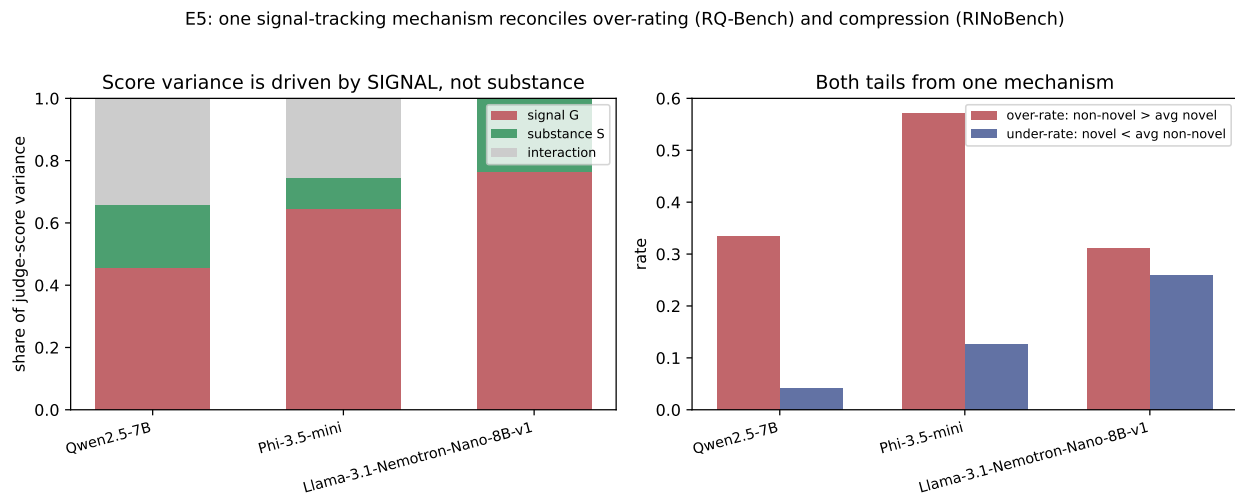


Figure 3: E5: signal dominates the score variance, producing both tails.

8 Discussion and limitations

The practical message is a warning: an LLM judge used as the verifier for automated research rewards sounding novel over being novel, the reward is farmable with substance fixed, and—unlike other judge biases—it cannot be cheaply steered out. **Limitations.** Small open judges and synthetic ideas; the pairwise attenuation for Qwen shows readout-dependence; near-perfect probe AU-ROC is partly a surface-lexical artifact, so the load-bearing mechanistic claim is the *failure* of the causal fix, not the detection; the reconciliation is internal (controlled) evidence, not a re-analysis of the RQ-Bench/RINoBench corpora; and we test difference-of-means steering, not fine-tuning. Whether frontier judges, natural ideas, and stronger interventions change the picture is open.

References

- [1] *On the Limits of LLM-as-Judge for Scientific Novelty Assessment* (RQ-Bench), arXiv:2606.12071.
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- [10] *Steering Large Language Models to Evaluate and Amplify Creativity*, arXiv:2412.06060.